- Which is worth more: a 1 into 4 7% receiver or a 1 into 4 8% receiver?
- Which is worth more: a 1 into 4 7% payer or a 1 into 4 8% payer?[4]

Now try these two:

- Which is worth more: a 1 into 4 7% receiver or a 2 into 3 7% receiver?
- Which is worth more: a 1 into 4 7% payer or a 2 into 3 7% payer?[5]

5.1.1 The Value of Swaptions at Expiration

The value at expiration of a European receiver swaption, which gives the holder the right to receive fixed in the underlying swap, is given by

$$\max(B_{\text{fixed}} - B_{\text{floating}}, 0), \tag{5.1}$$

where B_{fixed} is the value of the fixed leg of the swap including the fictitious principal at expiry, and B_{floating} is the value of the floating leg of the swap including fictitious principal at expiry.[6] Recall that the value of the floating side must be worth par on any reset date. The underlying swap is effective at the time of expiry and thus has its first reset at expiry. Thus, the value of the floating bond, B_{floating}, is par at expiry. The strike of the swaption determines the coupon of the fictitious fixed rate bond, and thus the value of the fixed rate bond, B_{fixed}, can be above, below, or equal to par at expiry.

From Equation 5.1, we see that the receiver swaption will be exercised when $B_{\text{fixed}} \geq B_{\text{floating}}$. This means that the receiver swaption will be exercised whenever B_{fixed} at expiry is worth more than par. This is equivalent to saying that the holder of a receiver swaption will exercise at expiry if the strike is greater than the then-prevailing swap rate for the corresponding maturity swap.[7] For example, in the case of a 1 into 4 7% receiver swaption, the swaption holder will exercise if the 4-year swap rate prevailing one year from now is less than 7%. The payoff from the swaption at expiry will depend

[4] The receiver struck at 8% must be worth more than an otherwise identical receiver struck at 7% since it is always better to receive more than less. Likewise, the 7% payer is worth more than the otherwise identical 8% payer since it is always better to pay less than more.

[5] The problem set at the end of this chapter will give you an opportunity to consider these two further.

[6] Of course, as we discussed in Chapter 3, the two counterparties do not literally exchange the fictitious principal at the maturity of the swap, but it does not change the value of the swap if we assume they do, and it is often convenient to assume so.

[7] At expiry, the then-prevailing swap rate will define a fictitious bond that is worth exactly par. Thus, any bond with a coupon greater than the then-prevailing swap rate will be worth more than par.

on where 4-year swap rates are in one year and how that compares to the strike of 7%.

Similarly, for a payer swaption, which gives the holder the right to pay fixed in the underlying swap, the value at expiry is given by

$$\max(B_{\text{floating}} - B_{\text{fixed}}, 0).$$

In the case of a payer swaption, the holder will exercise if the strike is less than the then-prevailing swap rate for the corresponding maturity swap. For example, in the case of a 1 into 4 7% payer swaption, the holder will exercise if the 4-year swap rate prevailing one year from now is greater than 7%. A receiver swaption is therefore a bullish instrument, increasing in value as the market rallies, whereas a payer swaption is a bearish instrument, increasing in value as the market sells off.

5.1.2 Swaption Parity

Notice that the following equation is true:[8]

$$\max(B_{\text{fixed}} - B_{\text{floating}}, 0) - \max(B_{\text{floating}} - B_{\text{fixed}}, 0) = B_{\text{fixed}} - B_{\text{floating}}.$$

In words, this means that, at expiration,

$$\begin{aligned}\text{Value of Receivers} - \text{Value of Payers} = {} & \text{Value of Receiving Fixed} \\ & \text{in Underlying Swap,}\end{aligned}$$

and since the payoffs from these positions hold with equality at expiration, the value of the positions today must be equal as well:

$$\begin{aligned}&\text{Value of Receiver} - \text{Value of Payer} \\ &\quad = \text{Value Today of Receiving Fixed in} \\ &\qquad \text{Underlying Forward Starting Swap}\end{aligned} \tag{5.2}$$

Equation 5.2 is referred to as *swaption parity*. It applies to a payer and receiver with the same expiry, the same strike, and the same underlying forward starting swap. It says that the value of the receiver less the value of the payer must be equal to the value of receiving fixed in the underlying forward swap that has a fixed coupon equal to the strike. For example, the value

[8] This equation says that $\max(A - B, 0) - \max(B - A, 0) = A - B$. To show that this is true, consider two cases. First, suppose that $A \geq B$. Then $\max(A - B, 0) = A - B$, and $\max(B - A, 0) = 0$, which means $\max(A - B, 0) - \max(B - A, 0) = A - B$. On the other hand, if $A < B$, then $\max(A - B, 0) = 0$, and $\max(B - A, 0) = B - A$, which means $\max(A - B, 0) - \max(B - A, 0) = -(B - A) = A - B$.

today of a 1 into 4 7% receiver less the value today of a 1 into 4 7% payer must be equal to the value today of receiving 7% fixed in a 4-year swap, one year forward. Suppose that we were to solve for the strike such that the receiver and the payer have the same value today. What is the significance of this strike?[9]

5.1.3 Uses of Swaptions

There are many ways in which investors use swaptions. Here are just a few:

- A manager of a mortgage portfolio may be concerned about a rally in 10-year swap rates over the next year that would trigger prepayments and cause the manager to reinvest at lower prevailing rates. The manager may buy 1 into 10 receiver swaptions to hedge against this risk.

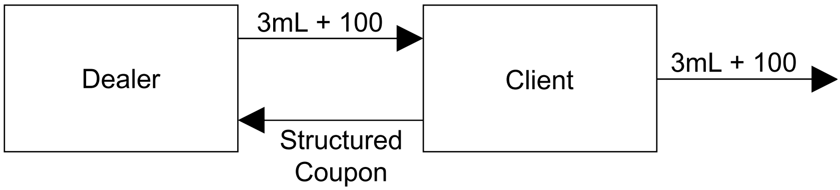
- A hedge fund believes that 5 into 10 swaption vol is going to go down in the near future. The hedge fund may sell some 5 into 10 swaption straddles. If, after a short time, implied vol on 5 into 10 swaption straddles declines, the hedge fund will be able to buy the straddles back at a lower price than which it sold them, recognizing a profit.

- Suppose that a corporation has a \$100mm floating rate liability over the next five years equal to 3-month LIBOR + 100 bps. The first quarterly setting on the liability is today, and LIBOR will be reset each quarter until the liability matures in five years.

  Assume that the current market environment is as in Table 1.1. Also assume that today's 3-month LIBOR setting is 0.38428%, and as such the first coupon due to be paid by the corporation in three months is 0.38428% + 100 bps = 1.38428%. Also note that today's 5-year (mid-market) swap rate is 1.627%. It can be shown that the 5-year swap rate on a quarterly money basis is approximately 1.601%, using Equations 1.8 and 1.9. Suppose further that the price of a 3-month into 4.75 year receiver swaption struck at 1.83% quarterly money is 123.1 bps upfront.

  The corporation is concerned about an increase in rates, which would raise its funding costs. Consider the following trade proposed to the corporation by a dealer:

[9] This strike must be equal to the par forward starting swap rate. Recall that the par swap rate in a forward starting swap is that rate that gives the swap zero value. If the payer and the receiver have the same value, swaption parity tells us that the underlying forward starting swap must have zero value, which means that the fixed rate in the swap is equal to the par forward starting swap rate.



where
Structured Coupon = 3mL + 75 for first three months
3mL + 75 or 2.58% thereafter,
at the option of the dealer in 3 months

Figure 5.1: Swaption Embedded in Funding Trade

1. The dealer will pay 3-month LIBOR + 100 bps to the corporation for five years.
2. The corporation will pay 3-month LIBOR + 75 bps to the dealer for five years.
3. The dealer has the one-time right in three months to convert the cash flows paid by the corporation to 2.58% fixed quarterly money for the remaining 4.75 years of the trade. If this right is not exercised by the dealer, then the corporation will continue to pay 3-month LIBOR + 75 bps for the remaining 4.75 years of the trade.

What's going on here? Refer to Figure 5.1, which depicts the floating liability of the corporation in conjunction with the trade propped by the dealer. If the swap dealer were to pay 3-month LIBOR + 100 bps to the corporation in return for 3-month LIBOR + 75 bps, it would lose 25 bps running over the five years. Using a proprietary swap model, the present value of 25 bps running over the five years on $100mm notional is $1,231,000 bps, or 123.1 bps upfront.

This loss is offset by the third leg of the trade outlined above. The one-time right that the dealer has to convert the cash flows paid by the corporation to 2.58% quarterly money is actually a receiver swaption. Under the terms of the trade, the corporation will be paying 3-month LIBOR + 75 for five years. However, in the third leg of the trade the dealer is essentially buying $100mm of a 3-month into 4.75-year receiver swaption, which gives the dealer the right to receive 2.58% versus 3-month LIBOR + 75. But receiver swaptions (and payer swaptions) are typically quoted such that the underlying swap pays LIBOR flat. That's not a problem however, because the value of this receiver is the same as a 3-month into 4.75-year receiver struck at 2.58% − 75 bps = 1.83% versus 3-month LIBOR flat (if these two receiver swaptions did not have the same value, there would be an arbitrage).[10] As mentioned above, the value of this receiver is 123.1 bps upfront, which offsets the loss the dealer locks in by paying away the 25 bp annuity.

Is this a good trade for the corporation? As stated, the corporation believes that rates are going to rise. It has sold a receiver swaption, which is an option that comes into the money as rates fall. If the corporation is correct and rates rise, the corporation will be paying 3-month LIBOR + 75, saving 25 bps running versus the LIBOR + 100 it would otherwise pay on its liability. However, if rates fall enough, the dealer will exercise the swaption, and the corporation will wind up paying 2.58% for the remaining 4.75 years of the trade. In the event the dealer exercises its receiver swaption, the corporation may well be paying an above-market fixed coupon, depending upon how much the market rallies. But even in this case, the corporation is doing better than if it had initially swapped its floating rate liability to fixed at a quarterly money rate of approximately 1.601% + 100 bps = 2.601%.[11]

5.1.4 Valuing Swaptions Using Black's Formula

Consider a T into n payer swaption struck at X, the right to pay a fixed rate X and receive LIBOR in a swap that lasts n years, starting in T years. Assume, for simplicity, that each side of the swap pays m times per year. Black's formula is often used to value European swaptions. It is assumed that the underlying forward swap rate (i.e., the n-year swap rate, T years forward) is

[10] Suppose the receiver swaption with the 1.83% strike versus LIBOR flat underlying swap were worth more than the receiver with the 2.58% strike versus LIBOR + 75 underlying. Then we would sell the former and buy the latter, pocketing money today. If at expiry our short were exercised against us, we would exercise our swaption, which would result in net zero cash flows. If our short were not exercised against us, we are guaranteed nonnegative cash flows. This would be an arbitrage, and thus the latter must be worth at least as much as the former. A similar argument shows that the former must be worth at least as much as the latter. In general, if we have a swaption in which the underlying swap has a spread to the LIBOR side, it is worth the same as a swaption in which we adjust the fixed rate accordingly and the floating side is LIBOR flat.

[11] The corporation could receive 3-month LIBOR quarterly money for five years in exchange for a fixed rate of 1.627%, the 5-year swap rate semi bond. This rate equates to a quarterly money rate of approximately 1.601%. If the corporation wanted to receive 3-month LIBOR + 100 bps quarterly money, the fixed rate it would have to pay would be 1.601% + 100 bps = 2.601% quarterly money.

lognormally distributed with an initial rate equal to F and volatility denoted by σ. Denote by R the (uncertain) n-year swap rate that will be observed at time T.[12]

The payer swaption will be exercised at time T if the n-year swap rate at time T, R, is greater than the strike X. Of course, if exercised the cash flows in the underlying swap are

$$\frac{\text{Notional}}{m} \times [L(t_i, t_{i+1}) - X] \tag{5.3}$$

that will be received m times per year for n years at times $t_i = T+1/m, T+2/m, \ldots, T+mn/m$, where $L(t_i, t_{i+1})$ represents the (uncertain) LIBOR rate that sets at time t_i. The payoff from the swaption is equivalent to a sequence of cash flows equal to

$$\frac{\text{Notional}}{m} \times \max(R - X, 0) \tag{5.4}$$

also received m times per year for n years at times $t_i = T + 1/m, T + 2/m, \ldots, T + mn/m$.[13]

Let $t_i = T+i/m$. According to Black's formula, the value of the swaption associated with the cash flow that may be received at time t_i due to exercise of this option is given by

$$\frac{1}{m} \times DF(0, t_i) \times [FN(d_1) - XN(d_2)], \tag{5.5}$$

where

$$d_1 = \frac{\ln(F/X) + \sigma^2 T/2}{\sigma\sqrt{T}} \tag{5.6}$$

[12] Suppose that we are thinking of pricing up a 5-year into 10-year swaption. The underlying swap rate is the 10-year swap rate, five years forward. We can bootstrap the swap curve and compute this rate today and denote its initial value by F. Tomorrow the underlying will be the 10-year swap rate, four years and 364 days forward (assuming it's not a leap year). We won't know the value of this variable until tomorrow, when we can bootstrap the curve and figure out what that forward rate is. The day after that the underlying will be the 10-year swap rate, four years and 363 days forward, and so on. Each and every day the forward rate is going to change (actually, the forward rate can change at any moment in time). It is assumed that the distribution of this forward rate at expiry in five years is lognormal with volatility σ. We can play this game each and every day, but the one thing we do know is that the forward rate will have to converge to the spot 10-year swap rate in five years, an uncertain rate that we said we would denote by R.

[13] What is the connection between Equations 5.3 and 5.4? This comes from our old trick that we learned in Section 3.3.1: the value of paying X and receiving LIBOR in a swap is the same as the sum of the value of (1) paying X and receiving LIBOR in a swap and (2) receiving R and paying LIBOR in a swap, since the latter of these transactions has zero value. The net position of (1) and (2) is a trade in which we are paying X and receiving R.

and

$$d_2 = d_1 - \sigma\sqrt{T}, \tag{5.7}$$

where once again F is the forward swap rate (i.e., the n-year swap rate, T years forward), σ is the volatility of this forward swap rate, and $DF(0, t_i)$ is the discount factor for time t_i. This gives us only the value of the cash flow associated with time t_i. The total value of the swaption is

$$\sum_{i=1}^{mn} \frac{1}{m} \times DF(0, t_i) \times [FN(d_1) - XN(d_2)]\,. \tag{5.8}$$

Let A be the annuity factor. That is, A is that number which when multiplied by \$1 gives the present value of a \$1 annuity per annum, paid in equal installments of $1/m$, m times per year, received at each time t_i, for $i = 1, \ldots, mn$. Then we can write that the value of the payer swaption is

$$A \times [FN(d_1) - XN(d_2)]\,, \tag{5.9}$$

where

$$A = \frac{1}{m} \sum_{i=1}^{mn} DF(0, t_i).$$

Similarly, the value of a T into n receiver swaption is

$$A \times [XN(-d_2) - FN(-d_1)]\,. \tag{5.10}$$

In order to find the value of a payer or receiver swaption in dollars, we would simply multiply the expressions in Equations 5.9 and 5.10, respectively, by the notional of the trade.

5.1.5 Swaption Vol

The volatility used to price a particular swaption depends on it expiry, the term of the underlying swap (also referred to as the *tail*), and the strike. If a swaption has a strike equal to the forward swap rate of the underlying swap, then the swaption is referred to as being struck *at-the-money forward* (often abbreviated "ATMF"). For example, in Section 3.3.4, we solved for the 1x4 swap rate (i.e., a 3-year swap rate, one year forward) in our stylized example and found a rate of 5.644%. Thus, a 1 into 3 swaption struck at 5.644% is referred to as being struck at-the-money forward.

Implied swaption vol refers to the swaption vol that is computed from a pricing model such as the one in Equation 5.9 or 5.10. That is, given an option price, implied vol refers to the value of vol input into a pricing model that results in the given option price. When we generically speak about implied swaption vol, we refer to volatilities for swaptions that are struck at-the-money forward. For the at-the-money forward strike, a swaption must

		Underlying Swap (Tail)								
		1	2	3	5	7	10	15	20	30
	1m	40.0	39.4	36.0	31.4	27.3	23.4	19.4	17.4	15.8
	3m	39.0	38.9	35.7	31.2	27.4	23.7	19.7	17.7	16.3
E	6m	39.3	37.2	33.9	30.0	26.5	23.2	19.4	17.5	16.4
x	1y	35.8	33.1	30.6	27.5	24.9	22.1	18.6	16.9	16.0
p	2y	30.4	28.2	26.5	24.2	22.2	20.1	17.2	15.8	14.9
i	3y	26.1	24.7	23.5	26.7	20.3	18.5	16.1	14.8	13.9
r	4y	23.3	22.3	21.3	19.9	18.7	17.2	14.9	13.9	13.1
y	5y	21.4	20.5	19.6	18.4	17.3	16.1	13.9	12.9	12.2
	7y	18.2	17.6	17.0	15.9	15.1	14.1	12.5	11.6	11.1
	10y	14.9	14.4	13.9	13.0	12.5	11.8	10.6	9.9	9.6

Table 5.1: Lognormal Implied Swaption Vol Matrix

still be identified by specifying its expiry and its tail. It is typical to show the implied volatilities of swaptions of various expiries and tails, all struck at-the-money forward, in a table referred to as a *vol grid* or *swaption matrix*.

Table 5.1 shows an example of such a matrix. In this vol grid, the expiry of the option is found along the first column of the matrix, and the tail (i.e., the maturity of the underlying swap) is found along the top row of the matrix. For instance, 1 year into 10-year swaptions have an implied volatility of 22.1%. It is important to note that this implied volatility applies only to 1 year into 10-year swaptions that are struck at-the-money forward. 1 year into 10-year swaptions that have different strikes may (and most often do) trade at different implied volatilities. Short-dated swaptions (considered by many to be those with time until expiration of a year and under) are referred to as *gamma*, whereas longer-dated options are referred to as *vega*.[14]

5.1.6 Pricing Swaptions in Our Stylized Example

We can apply Black's formula to price swaptions in the context of our stylized example in which we have fixed annual bond versus 1-year LIBOR. This amounts to setting $m = 1$ in Equation 5.9 and Equation 5.10 and recognizing that the discount factors we need are formed from the zero coupon rates obtained from bootstrapping the swap curve. Upon exercise, the underlying swap has cash flows one time per year for n years at times $T+1, T+2, \ldots, T+n$. In our stylized example, Black's formula as per Equation 5.9 for the value of a T into n payer swaption struck at X reduces to

[14]Short-dated options tend to have high gamma and low vega, whereas long-dated options tend to have low gamma and high vega, attributing to these choices of names.

$$A \times [FN(d_1) - XN(d_2)], \tag{5.11}$$

where

$$d_1 = \frac{\ln(F/X) + \sigma^2 T/2}{\sigma\sqrt{T}} \tag{5.12}$$

and

$$d_2 = d_1 - \sigma\sqrt{T}, \tag{5.13}$$

and

$$A = \sum_{t=T+1}^{T+n} DF(0,t) \tag{5.14}$$

where

$$DF(0,t) = \frac{1}{[1 + \mathrm{zc}(t)]^t}. \tag{5.15}$$

Similarly, the value of a T into n receiver swaption struck at X is

$$A \times [XN(-d_2) - FN(-d_1)]. \tag{5.16}$$

As a reminder, F is the forward swap rate (i.e., the n-year swap rate, T years forward), σ is the implied volatility of this forward swap rate, and $\mathrm{zc}(t)$ is the zero coupon rate for time t. The sum in Equation 5.14 represents the annuity factor: that number which, when multipled by \$1, gives the value of a \$1 annuity received at the end of each year in the underlying swap. In order to determine the premium of a payer and receiver swaption in dollars, we would multiply the expressions in Equations 5.11 and 5.16, respectively, by the notional of the trade.

Example

Let's consider our stylized example and price up a 1 into 3 payer swaption struck at 5.644%. Assume that the swap curve is as in Table 5.2 and assume 1 into 3 implied vol is 20%. An input in the pricing of the swaption is the calculation of the underlying forward swap rate. In Section 3.3.4 we showed that the 1x4 swap rate (i.e., the 3-year swap rate, one year forward) was equal to 5.644%. Thus, this 1 into 3 payer swaption is struck at-the-money forward, and we have $F = X = 5.644\%$. Figure 5.2 demonstrates how we price up the payer swaption. The value of the swaption is around 115.5 bps. For a

Maturity	Treasury Yield	Swap Spread	Swap Rate (Ann 30/360)
Year 1	4.80%	20	5.00%
Year 2	4.88%	23	5.11%
Year 3	5.05%	25	5.30%
Year 4	5.19%	28	5.47%
Year 5	5.25%	35	5.60%

Table 5.2: The Swap Curve

Swaption model		1	into	3		
Time	Swap rate	zc	Period	$DF(0,t)$		
1	5.000%	5.000%	0x1		d_1	0.1000
2	5.110%	5.113%	1x2	0.9051	d_2	-0.1000
3	5.300%	5.312%	2x3	0.8562		
4	5.470%	5.494%	3x4	0.8074	$FN(d_1)$ - $XN(d_2)$	0.004496
					Payer swaption	1,154,863
					(in bps)	115.5
F	5.644%		A =	2.5687		
σ	20.0%					
T	1		Notional	100,000,000		
X	5.644%					

Figure 5.2: Pricing Up the 1 into 3 Payer Swaption

notional of \$100mm, the value of the 1 into 3 payer swaption is \$1,154,863.[15] Similarly, we can price up a 1 into 3 receiver swaption at the same strike of

[15] As is the case in the market for caps and floors, it is common in the swaption market to round the upfront cost to either the nearest tenth of a basis point or quarter of a basis point. The premium in dollars, obtained by multiplying the cost of the option in bps by the notional, has been calculated to the nearest dollar. However, in practice the premium is often given in some round amount. In this example, the premium might be rounded to \$1,155,000. The benefit of talking about premia in terms of upfront bps rather than dollars is that the upfront bps is invariant to a change in notional (assuming the aggressiveness of the price does not vary due to a change in the size of the trade). For example, for a notional of \$200mm, the price of the swaption is still 115.5 bps upfront, or 115.5 bps $\times$ \$200mm = \$2,310,000, twice the dollar cost of a \$100mm trade.